When did living land plants arise? What their fossil age limits require

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Abstract

Molecular clocks place the ancestor of living land plants anywhere from the Cambrian to the Late Ordovician, partly because they use different fossil maximum ages. We examined three maxima: 515.5 and 459.1 million years ago (Ma) for living land plants, and 431.2 Ma for living vascular plants (ferns, seed plants and relatives). We audited their sources, modelled Libyan spore records and bounded when the vascular-plant body-fossil record began. The oldest maximum rests on a qualitative preservation argument. In Libya, three-rayed (trilete) spores occur in 51 of 59 younger rock samples and none of 28 older ones; with equal delivery, constant local spore input from 459.1 Ma fits at up to 0.091 of the younger level. For vascular-plant bodies, constant recording gives a one-sided 95% older bound of 428.8 Ma on their record's start, and alternative counting units give 432.4–434.05 Ma. If recognisable bodies appeared with the ancestor, ancestors at 444–452 Ma fit when earlier bodies were recorded 8–12 times less often, or 2.4–4.5 times under alternative units. Older dates for living land plants fit the fossil record if early plants were found several times less often, or recognisable features evolved later.

Keywords: molecular clock; fossil calibration; land plants; vascular plants; stratigraphic confidence interval; Ordovician–Silurian

Introduction

Molecular clocks estimate when lineages diverged by combining DNA differences with fossil limits on the ages of selected ancestors. A crown group comprises the most recent ancestor shared by its living members and all its descendants, including extinct lineages. For the ancestor of all living land plants (crown embryophytes), Morris et al. [1] obtained 515–474 Ma across seven tree topologies and Harris et al. [2] 515–493 Ma. Hooper et al. [3] reanalysed Harris et al.'s DNA data with revised fossil calibrations and obtained 457–450 Ma, a mean 55 Myr younger. For the ancestor of living vascular plants (crown tracheophytes: clubmosses and their relatives, ferns and seed plants), the three studies give 451–419, 452–447 and 432–428 Ma (figure 1); Su et al. [4] obtained older land-plant ages still, under a Proterozoic maximum. The dates bear on the order in which plant characters were assembled, and on how early plants changed productivity, weathering and climate [5–8]. They date the ancestors of living plants; extinct lineages may have reached land earlier.

A fossil sets a minimum age: its lineage existed by the age of its oldest secure member. An absence-based maximum adds an argument that older rocks would probably have preserved and yielded recognisable remains had the lineage existed [9]. How calibrations are specified can strongly shape estimated dates, and with large sequence datasets they can dominate the uncertainty in absolute ages [10,11]. A soft maximum permits older ages through a specified tail of the calibration prior [12]. Whether absence can set a maximum is contested [13,14], and best practice asks each calibration to justify its maximum from the relevant geological, geographical and taphonomic record [15].

Linking a fossil to an ancestor crosses two intervals: from divergence to the first recognisable character, and from that character to its first recorded fossil [16]. Both are uncertain for early land plants. Dispersed spores appear tens of millions of years before branched plant bodies [17–19], terrestrial deposits are scarce before the latest Silurian, at least in western Europe [5], and the oldest parent plants known in detail are minute [20].

Three maxima divide the current clocks (table 1). Morris et al. set 515.5 Ma for living land plants, arguing that older marine rocks rich in algal cysts would also have yielded land-plant spores [1,14]. Hooper et al. set 459.1 Ma from the oldest possible trilete (three-rayed) spores, in the Borenshult-1 core of southern Sweden [21], and 431.2 Ma for living vascular plants from Irish *Cooksonia* [22]. The three therefore rest on a preservation analogy, on dispersed spores and on plant bodies, each with its own route into the record, and each needs its own test.

We ask three questions of the published record. What evidence supports each maximum, and which links connect it to the dated ancestor? How much sustained spore input before the Silurian do the records of one Libyan basin allow? What reduction in early fossil recording, or delay before recognisable bodies, would accommodate older vascular-plant ancestors? Established theory sets an expectation for each. An absence argument needs documented opportunities for delivery, preservation and discovery [15,16]. With comparable exposure, accumulating zero records increasingly constrain high sustained spore input. For a fixed oldest find, younger cutoff and recording history, more independent finds narrow the bound on when a record began, and lower earlier recording extends it [23,24]. The audit identifies what each maximum assumes; the quantitative tests bound local spore input and describe the earlier recording or diagnostic delays compatible with older vascular-plant dates.

Table 1. The three maxima examined: the evidence each rests on, the test the published record allows, and the links a clock must still assume. GTS2012, Geologic Time Scale 2012.

Maximum	Set by	Evidence	Test available	Links assumed
515.5 Ma, living land plants	Morris et al.	Base of the *Olenellus* Biozone, below the oldest possible Cambrian spore-like packets; older marine rocks with algal cysts	Affinity of the Cambrian packets; any enumerated older sequence	Spore delivery to marine basins, catchment, search effort, spore production by the earliest members

Maximum	Set by	Evidence	Test available	Links assumed
459.1 Ma, living land plants	Hooper et al.	GTS2012 Sandbian base plus uncertainty, keyed to the oldest possible trilete spore (Borenshult-1, Sweden)	Older zeros against younger positives, Murzuq Basin, Libya	Earliest crown members produced triletes, the trait persisted, producers lived in the catchment
431.2 Ma, living vascular plants	Hooper et al.	GTS2012 Homerian base plus uncertainty, keyed to Irish Homerian *Cooksonia*	A bound from the body-fossil record, and what older ancestors require	Recording before the oldest find, delay before recognisable bodies, the anchor lies within the vascular crown

Material and methods

(a) Timescale and readings. Analysed ages use the stage bases of the International Chronostratigraphic Chart, v2024/12 [25]; literature ages were mapped by stage. Published maxima keep their source timescale, GTS2012, so they can be compared as their authors set them. Each analysis declares its reading of disputed identifications, main or inclusive, so contested fossils enter as alternatives.

(b) Source audit. For each maximum we traced the calibration fossil, its age rule and its recovery route to the original descriptions, and recorded each published reading of the fossil's identity with its evidence [15]. A reported non-detection applies to the material examined, and a missing report counts as a zero only where the source reports the target fully.

(c) Libyan spore records. Abuhmida [26] reported spores and other microfossils from Ordovician to lower Silurian wells in the Murzuq Basin, southwest Libya. We transcribed all 164 appendix samples with their chart and prose reports. The primary frame uses the chart-coded cuttings (rock chips returned during drilling) of five wells: three supply 28 cuttings older than the Llandovery (earliest Silurian), and all five supply 59 Llandovery reference cuttings. The source states the absence of trilete spores explicitly for part of the sequence; for the Hawaz Formation, zeros rest on complete reporting charts and the stated microscopy protocol. The basin lay near the Ordovician South Pole, within the area of Hirnantian glaciation [27,28] (figure 3*a*).

(d) Spore encounter model. We jointly fitted the reference encounter rate and older input, with a mean-one gamma-distributed effect (shape and rate 2) shared by the cuttings of each well. Given this effect and their ages, which are uniform within each cutting's stage range, cutting outcomes are independent. Older input is a fraction f of Llandovery input, and a fixed delivery ratio ρ covers differences in transport, preservation and preparation. We computed upper 95% compatibility limits on f for constant input, and for the exposure-weighted mean of variable histories, from a postulated start, with input between zero and the reference level. The potential screen sets older reports to zero and checks whether the frame could exclude constant reference-level input when every older cutting is exposed, with a usable younger reference. Each simulation uses the same screen; the analysis uses the actual reports. Acceptance simulations independent of calibration then qualified each declared quantity on a finite scenario grid. Among replicates passing the screen, with withheld claims counted as non-errors, the one-sided 95% lower confidence limit on coverage had to exceed 0.93; borderline quantities went to a preplanned pooled rule (electronic supplementary material, S2).

(e) Body-fossil units. We assembled land-plant body fossils dated to 453–393 Ma from Paleobiology Database (PBDB) occurrences, whose names an AI model classified with cited sources before our audit, and from a source-reviewed set of the oldest records with their competing readings. Collections, taxa or papers

sharing a locality and age count once, as proxies for independent fossil horizons. We declare three nested categories. Land-plant bodies are any accepted macro- or mesofossil. Polysporangiophytes have documented branched axes bearing sporangia (spore capsules), or placement within the vascular crown. Crown vascular plants belong to the lycophyte line (clubmosses and relatives) or the euphyllophytes (ferns and seed plants) [29–32]. The main reading gives 78, 71 and 50 units, of which 76, 69 and 48 are certainly older than the end of the Emsian (393.47 Ma) (electronic supplementary material, S5.1).

(f) Bound on record start. Let $w(s)$ be a fixed relative rate at which identifiable occurrences of a category are recorded at age s, and $U(a) = \int_c^a w(s)\,ds$ the cumulative recording above a younger cutoff c, here the end of the Emsian. For n independent units with oldest age A, the one-sided 95% older bound T on the start of the recognisable record solves $U(T) = U(A)\,0.05^{-1/n}$ (figure 2). With constant w this is the classical confidence interval on a stratigraphic range, and the absolute rate cancels [23,24]; varying w follows the extensions to non-random fossil horizons [33,34]. Units dated by ranges enter through the greatest possible oldest age and the minimum number guaranteed to lie within the window. The bound concerns the start of the record; the Discussion treats its link to ancestors.

(g) Recording histories and shape tests. We compared six fixed histories: constant recording; recording proportional to PBDB land and marginal-marine collections per Myr [35], with or without collections matching plant occurrences; exponential declines into the past with 25 or 10 Myr timescales; and plant-excluded collection density multiplied by the 10 Myr decline. The collection records extend to 469.4 Ma, which sets the analysis domain. Finite-sample Kolmogorov–Smirnov tests compared each history with the observed age sequence, conditional on the count and the oldest find. For each history we searched every admissible allocation of ages within their ranges, and each inclusion state of the units straddling the cutoff, for the most favourable allocation [compare 36]. A history that fails under every allocation and state is rejected for this diagnostic (electronic supplementary material, S5.2).

(h) Recording contrast and diagnostic delay. The primary step model keeps later recording constant and lowers earlier recording before 426.7 Ma, the oldest possible age of a crown vascular-plant find. For a candidate ancestor age t and an assumed delay d before recognisable bodies, the earlier-to-later rate ratio that places $t - d$ on the bound is $r = (A-c)(0.05^{-1/n}-1)/(t-d-A)$, defined where $t-d > A$; the recording contrast is $1/r$. We placed the step after inspecting the record, so its contours describe retrospective compatibility. Longer assumed delays reduce the contrast a given candidate age requires [16].

Results

(a) The 515.5 Ma maximum rests on a qualitative control. Morris et al. placed this maximum below the oldest possible total-group records, spore-like packets and clusters from the Cambrian Stage 4 Rome Formation, at the base of the *Olenellus* Biozone (about 515.5 Ma). Its preservation control, older marine sequences with algal cysts and without land-plant spores, shows that walled microfossils survived and were recovered [1,14]. Spore delivery to those basins, the land areas they drained and the search effort remain unquantified in the sources we assessed. All three sources we examined assign the Cambrian packets to algae or pre-land-plant lineages: Charophyta [37], a pre-embryophytic grade [38] or ancestral aeroterrestrial algae [39] (electronic supplementary material, S4).

(b) The younger maxima follow a stage-base rule. Hooper et al. set both younger maxima as a GTS2012 stage base plus its uncertainty. They keyed 459.1 Ma (458.4 + 0.7 = 459.1 Ma) to a possible trilete spore at 48.2 m in the Borenshult-1 core [21]. The oldest definite trilete level, at 43.1 m, is also Sandbian and lies below the 454.41 Ma Kinnekulle K-bentonite, so the rule gives the same value from either level (electronic supplementary material, S2.4). They keyed 431.2 Ma (430.5 + 0.7 = 431.2 Ma) to Irish Homerian *Cooksonia* [22]. The older Czech *Cooksonia barrandei*, a branched sporophyte with terminal sporangia from a Sheinwoodian (early Silurian) horizon inferred from slab fauna and correlation [30], meets the same character criterion (electronic supplementary material, table S13). On the current timescale the rule gives outer values of 431.9 Ma for the Homerian and 434.1 Ma for the Sheinwoodian.

(c) Libyan zeros limit sustained, well-delivered spore input. Trilete spores occur in 51 of 59 reference cuttings and none of the 28 older cuttings (figure 3*a*). For input starting at 459.1 Ma with equal delivery, the upper compatibility limit is 0.091 of reference input for constant histories, and 0.184 for the exposure-

weighted mean of bounded variable histories (figure 3*b*; electronic supplementary material, table S12). With delivery at a tenth of the reference, constant input near the reference level fits (0.910). For constant histories the record constrains the product ρf, which stays at 0.091 across delivery scenarios.

(d) The Libyan limits cover the later Ordovician. Most older exposure comes from 18 Katian–Hirnantian (Late Ordovician) cuttings, all younger than 459.1 Ma. Only 10 broadly dated Darriwilian (Middle Ordovician) cuttings, from two wells, have age ranges extending before 459.1 Ma. Their contribution under a start at that age depends on how ages are spread within the stage; shared midpoint ages give them none. The limits therefore concern sustained input through the later Ordovician in one high-latitude basin.

(e) The vascular-plant body record has an older bound near 429 Ma. The 48 crown vascular-plant units give a one-sided 95% older bound of 428.8 Ma under constant recording (figure 4*a*). Histories tracking collection density give 428.9 Ma, or 428.5 Ma with plant-matched collections removed, and a 25 Myr decline into the past gives 431.6 Ma. The pure 10 Myr decline leaves the start unresolved through 469.4 Ma and fails the shape diagnostic under every admissible allocation and inclusion state. For the 25 Myr decline, one admissible allocation of all 50 possible units gives $p = 0.067$, whereas every allocation of the 48 guaranteed units fails (best $p = 0.035$).

(f) Counting units move the bound across 431.2 Ma. Collapsing units to region–stage cells (19 units) gives 432.4 Ma, and the curated oldest-record set alone (15 units) gives 434.05 Ma, both under constant recording. Shifting stage boundaries by their quoted uncertainties, in the tested shared-boundary scenarios, moves the primary bound between 427.31 and 430.42 Ma (electronic supplementary material, S5.3).

(g) Older vascular-plant dates fit sparser early recording or later recognisable bodies. In the primary step model, halving earlier recording gives 431.0 Ma, still younger than the 431.2 Ma maximum. With zero assumed delay, ancestors at 444–452 Ma, spanning the Morris et al. mean dates and the Harris et al. interval, need earlier recording 8–12 times lower than later recording (figure 4*b*). Under region–stage or curated units the contrast falls to 2.4–4.5 times, and 431.2 Ma needs none. With constant recording, the same ancestors instead need assumed delays of at least 15–23 Myr before recognisable bodies. Smaller contrasts combine with shorter delays (figure 4*b*).

(h) One Ordovician find moves the broader land-plant bound. Sporangial fragments with in situ cryptospores from the Ghaba-1 borehole, Oman, dated to the Katian on later correlation [40], form one land-plant body unit. Adding them moves the broader category's constant-recording bound from 434.5 to 455.2 Ma, and the augmented sequence fails the shape diagnostic under all six fixed histories. The fragments came from palynological preparation of borehole material, whereas most later units are compressions collected from bedding planes [41].

Discussion

The early record allows older ancestors, and the analyses state what each would require. In the Murzuq Basin, sustained, well-delivered trilete input at the Llandovery level through the later Ordovician conflicts with the zero records; lower delivery, lower local input or producers outside the catchment all remain compatible. Under constant or collection-density recording, the one-sided 95% older bound on the recognisable crown vascular-plant record lies near 429 Ma, and coarser or curated units put it at 432.4–434.05 Ma. For these unit treatments, sparser earlier recording, later recognisable bodies or both can accommodate ancestors older than the corresponding fixed-history bounds; the models size each requirement in aggregate.

Several processes could have made early plants scarcer in the record. Producers may have been rarer, more geographically restricted, concentrated where little terrestrial sediment accumulated, or preserved in intervals that have been searched less. Body size, cuticle and lignified tissue could also raise the chance that a body is preserved, found and recognised. Branched sporophytes and conducting tissue were acquired stepwise along the lineages being dated [42]. The early branched sporophyte *C. barrandei* is preserved in marine sediments [30], and the Oman fragments show that small sporangia can survive [40]. The models quantify the combined contrast a candidate age requires; the contribution and plausibility of each process remain open.

Silvestro et al. [43] modelled genus-level PBDB plant occurrences with a joint Bayesian model of preservation and diversification, and estimated the origin of vascular plants at a mean of 433.5 Ma (95% highest posterior

density 449.0–424.0 Ma). They estimate when the earliest vascular-plant genus lineage arose, with a fitted, varying preservation rate; we bound the start of the recognisable record of a declared crown category under fixed recording histories, then ask what older ancestors require. The two analyses target different quantities from partly shared data and test different assumptions. CladeDate builds calibration densities directly from fossil times [44], and Hooper et al. link spore-character reconstruction to revised calibrations. We make the maxima's observational requirements explicit, quantifying local spore input and the body-recording contrasts or diagnostic delays compatible with older vascular-plant dates.

Conditions. The spore limits assume an indigenous younger reference, comparable reporting in older and younger cuttings, ages uniform within stages, and cutting outcomes independent given a shared well effect. Three exploratory dependence checks, with unqualified coverage, give contours of 0.0995 with separate reference rates per well and no random well effect, near 0.4 with well–formation block weighting, and 0.185–0.365 (exposure-weighted) with one well left out at a time. Linking the local spore record to ancestors requires that the earliest crown embryophytes produced recognisable trilete spores, that the trait persisted and that producers lived in the catchment (electronic supplementary material, S6).

The body bounds and conservative shape rejections assume independent continuous ages; fixed histories, cut-off, timescale, inventory and unit rules; and an observation model whose age ranges contain all true ages and inclusion states with probability one. Geological ranges have no demonstrated joint coverage; if containment fails with probability δ, the direct test-size bound is $\alpha+\delta$, where α is the exact-age significance level. Matched exact-age simulations gave coverage of 0.941–0.958, falling to about 0.78 when paired duplicate horizons were treated as independent (electronic supplementary material, S5.6). The step model is retrospective. Linking the vascular-body bound to ancestors requires crown membership of the oldest finds and a specified delay before recognisable bodies.

Published evidence could narrow these requirements. Records of searched Silurian terrestrial beds, with effort and empty beds included, would help constrain early recording, and compilations of rock units could supply proxies for the terrestrial rock available at each age [45]. A second complete older-zero, younger-positive spore sequence outside polar Gondwana, with counted target and control grains, would test whether the Libyan result is general; an enumerated marine sequence older than 515.5 Ma, with comparable recovery documented, would help quantify the oldest maximum's absence argument. Finer, independent ages for the Yea (Victoria, Australia) and Bathurst Island (Arctic Canada) horizons, which hold the oldest possible crown vascular-plant finds, would sharpen the body bound [29,31]. Characters placing the oldest branched sporophytes on the lycophyte or euphyllophyte side of the vascular crown could change which fossils enter the category, as well as their link to the ancestor.

Conclusion

An absence-based fossil maximum combines expectations about ancestral characters with the chance of finding their remains in older rocks. Clock builders can state each maximum's character, geographical and recording assumptions, and justify the probability assigned to older ages against them [12,15]. Testing alternative defensible maxima then shows how much each assumption moves the timescale, and sampling the joint age prior without molecular data, then comparing its effective marginal ages with the posteriors, shows what the molecular data add [46]. The same separation of evidence, recording and mapping applies to any maximum argued from absence. Older dates for living land plants fit the fossil record if early plants were found several times less often, or recognisable features evolved later.

Ethics

We analysed published descriptions and data. The study involved no fieldwork, specimen collection or research involving human participants.

Data accessibility

The repository https://github.com/BWBrook/silurian_plants holds the source-linked derived tables, model code, qualification evidence, body-record outputs with hash receipts, figure and table scripts, and the claims ledger that binds every reported number to its evidence. The frozen PBDB collections extract used by the

effort histories is archived under CC0 [35]. Copyrighted full texts are excluded from the repository; source routes and held-file hashes are recorded separately. Openly licensed third-party deposits are fetched by script and hash-checked. Supplementary material is available online.

Declaration of AI use

Claude Opus 5.5 (Anthropic) developed and ran analyses and drafted and revised the manuscript. OpenAI Codex (`gpt-6.1-sol` and `gpt-6.1-terra`) curated and checked source data, implemented analyses, audited statistical methods and reproduction, and revised the manuscript. Earlier Codex work used `gpt-5.6-sol` and `gpt-5.6-luna` for source recovery and methodological checks. Claude and Codex reviewed each other's work. GPT-6 Pro (OpenAI) supplied independent critiques, which were checked against sources and saved outputs before integration. Barry W. Brook originated and guided the programme and contributed scientific decisions; he will review the final manuscript before submission. Task-specific roles, model versions and the limits of sessions whose exact variant was unavailable are recorded in `docs/charter.md`, `docs/audits/` and `manuscript/reconstruction/`, with adopted decisions in `docs/decisions.md`.

Authors' contributions

[Byline and final CRediT wording remain deferred to the submission phase. A contribution inventory is prepared in `manuscript/reconstruction/4-submission-materials.md`.]

Conflict of interest declaration

[Barry to confirm.]

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References

1. Morris JL *et al.* 2018 The timescale of early land plant evolution. *Proceedings of the National Academy of Sciences* **115**. (doi:10.1073/pnas.1719588115)

2. Harris BJ, Clark JW, Schrempf D, Szöllősi GJ, Donoghue PCJ, Hetherington AM, Williams TA. 2022 Divergent evolutionary trajectories of bryophytes and tracheophytes from a complex common ancestor of land plants. *Nature Ecology & Evolution* **6**, 1634–1643. (doi:10.1038/s41559-022-01885-x)

3. Hooper J, Worth AM, Wellman CH, Clark JW, Donoghue PCJ. 2026 Reconstructing spore evolution recalibrates the timescale of land plant diversification. *Nature Plants* (doi:10.1038/s41477-026-02432-3)

4. Su D, Yang L, Shi X, Ma X, Zhou X, Hedges SB, Zhong B. 2021 Large-Scale Phylogenomic Analyses Reveal the Monophyly of Bryophytes and Neoproterozoic Origin of Land Plants. *Molecular Biology and Evolution* **38**, 3332–3344. (doi:10.1093/molbev/msab106)

5. Kenrick P, Wellman CH, Schneider H, Edgecombe GD. 2012 A timeline for terrestrialization: consequences for the carbon cycle in the Palaeozoic. *Philosophical Transactions of the Royal Society B: Biological Sciences* **367**, 519–536. (doi:10.1098/rstb.2011.0271)

6. Donoghue PCJ, Harrison CJ, Paps J, Schneider H. 2021 The evolutionary emergence of land plants. *Current Biology* **31**, R1281–R1298. (doi:10.1016/j.cub.2021.07.038)

7. Strother PK, Foster C. 2021 A fossil record of land plant origins from charophyte algae. *Science* **373**, 792–796. (doi:10.1126/science.abj2927)

8. Lenton TM, Crouch M, Johnson M, Pires N, Dolan L. 2012 First plants cooled the Ordovician. *Nature Geoscience* **5**, 86–89. (doi:10.1038/ngeo1390)

9. Benton MJ, Donoghue PCJ. 2006 Paleontological Evidence to Date the Tree of Life. *Molecular Biology and Evolution* **24**, 26–53. (doi:10.1093/molbev/msl150)

10. Ho SYW, Phillips MJ. 2009 Accounting for Calibration Uncertainty in Phylogenetic Estimation of Evolutionary Divergence Times. *Systematic Biology* **58**, 367–380. (doi:10.1093/sysbio/syp035)

11. dos Reis M, Donoghue PCJ, Yang Z. 2015 Bayesian molecular clock dating of species divergences in the genomics era. *Nature Reviews Genetics* **17**, 71–80. (doi:10.1038/nrg.2015.8)

12. Yang Z, Rannala B. 2005 Bayesian Estimation of Species Divergence Times Under a Molecular Clock Using Multiple Fossil Calibrations with Soft Bounds. *Molecular Biology and Evolution* **23**, 212–226. (doi:10.1093/molbev/msj024)

13. Hedges SB, Tao Q, Walker M, Kumar S. 2018 Accurate timetrees require accurate calibrations. *Proceedings of the National Academy of Sciences* **115**. (doi:10.1073/pnas.1812558115)

14. Morris JL *et al.* 2018 Reply to Hedges et al.: Accurate timetrees do indeed require accurate calibrations. *Proceedings of the National Academy of Sciences* **115**. (doi:10.1073/pnas.1812816115)

15. Parham JF *et al.* 2011 Best Practices for Justifying Fossil Calibrations. *Systematic Biology* **61**, 346–359. (doi:10.1093/sysbio/syr107)

280 16. Marshall CR. 2019 Using the Fossil Record to Evaluate Timetree Timescales. *Frontiers in Genetics* **10**. (doi:10.3389/fgene.2019.01049)

281 17. Wellman CH, Gray J. 2000 The microfossil record of early land plants. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences* **355**, 717–732. (doi:10.1098/rstb.2000.0612)

282 18. Rubinstein CV, Gerrienne P, de la Puente GS, Astini RA, Steemans P. 2010 Early Middle Ordovician evidence for land plants in Argentina (eastern Gondwana). *New Phytologist* **188**, 365–369. (doi:10.1111/j.1469-8137.2010.03433.x)

283 19. Steemans P, Hérissé AL, Melvin J, Miller MA, Paris F, Verniers J, Wellman CH. 2009 Origin and Radiation of the Earliest Vascular Land Plants. *Science* **324**, 353–353. (doi:10.1126/science.1169659)

284 20. Edwards D, Morris JL, Richardson JB, Kenrick P. 2014 Cryptospores and cryptophytes reveal hidden diversity in early land floras. *New Phytologist* **202**, 50–78. (doi:10.1111/nph.12645)

285 21. Rubinstein CV, Vajda V. 2019 Baltica cradle of early land plants? Oldest record of trilete spores and diverse cryptospore assemblages; evidence from Ordovician successions of Sweden. *GFF* **141**, 181–190. (doi:10.1080/11035897.2019.1636860)

286 22. Edwards D, Feehan J. 1980 Records of Cooksonia-type sporangia from late Wenlock strata in Ireland. *Nature* **287**, 41–42. (doi:10.1038/287041a0)

287 23. Strauss D, Sadler PM. 1989 Classical confidence intervals and Bayesian probability estimates for ends of local taxon ranges. *Mathematical Geology* **21**, 411–427. (doi:10.1007/bf00897326)

288 24. Marshall CR. 1990 Confidence intervals on stratigraphic ranges. *Paleobiology* **16**, 1–10. (doi:10.1017/s0094837300009672)

289 25. Cohen KM, Finney SC, Gibbard PL, Fan J-X. 2013 The ICS International Chronostratigraphic Chart. *Episodes* **36**, 199–204. (doi:10.18814/epiiugs/2013/v36i3/002)

290 26. Abuhmida FH. 2013 Palynological analysis of the Ordovician to Lower Silurian sediments from the Murzuq Basin, southwest Libya. PhD thesis, University of Sheffield. See https://etheses.whiterose.ac.uk/id/eprint/4475/.

291 27. Cocks LRM, Torsvik TH. 2021 Ordovician palaeogeography and climate change. *Gondwana Research* **100**, 53–72. (doi:10.1016/j.gr.2020.09.008)

292 28. Scotese CR, Song H, Mills BJW, van der Meer DG. 2021 Phanerozoic paleotemperatures: The earth's changing climate during the last 540 million years. *Earth-Science Reviews* **215**, 103503. (doi:10.1016/j.earscirev.2021.103503)

293 29. Kotyk ME, Basinger JF, Gensel PG, de Freitas TA. 2002 Morphologically complex plant macrofossils from the Late Silurian of Arctic Canada. *American Journal of Botany* **89**, 1004–1013. (doi:10.3732/ajb.89.6.1004)

294 30. Libertín M, Kvaček J, Bek J, Žárský V, Štorch P. 2018 Sporophytes of polysporangiate land plants from the early Silurian period may have been photosynthetically autonomous. *Nature Plants* **4**, 269–271. (doi:10.1038/s41477-018-0140-y)

295 31. Earp C. 2024 Notanopliid brachiopods and associated fossils from the Yea area, central Victoria: a key to resolving age and correlation problems in a complex geological setting. *Proceedings of the Royal Society of Victoria* **136**. (doi:10.1071/rs23021)

296 32. Libertín M, Kvaček J, Bek J. 2024 The genus Aberlemnia and its Silurian–Devonian fossil record. *Review of Palaeobotany and Palynology* **320**, 105017. (doi:10.1016/j.revpalbo.2023.105017)

297 33. Marshall CR. 1994 Confidence intervals on stratigraphic ranges: partial relaxation of the assumption of randomly distributed fossil horizons. *Paleobiology* **20**, 459–469. (doi:10.1017/s0094837300012938)

298 34. Marshall CR. 1997 Confidence intervals on stratigraphic ranges with nonrandom distributions of fossil horizons. *Paleobiology* **23**, 165–173. (doi:10.1017/s0094837300016766)

299 35. Brook BW. 2026 Frozen Paleobiology Database all-taxa collections extract: Darriwilian–Emsian, 28 October 2026. (doi:10.5281/zenodo.24028289)

300 36. Berger RL, Boos DD. 1994 *P* Values Maximized Over a Confidence Set for the Nuisance Parameter. *Journal of the American Statistical Association* **89**, 1012–1016. (doi:10.1080/01621459.1994.10476836)

301 37. Strother PK. 2016 Systematics and evolutionary significance of some new cryptospores from the Cambrian of eastern Tennessee, USA. *Review of Palaeobotany and Palynology* **227**, 28–41. (doi:10.1016/j.revpalbo.2015.10.006)

302 38. Taylor WA, Strother PK. 2024 Ultrastructure of Cambrian cryptospores and the early evolution of the plant spore wall. *Fossil Imprint* **80**, 90–106. (doi:10.37520/fi.2024.009)

303 39. Wang K, Xu H-H, Yin L-M. 2022 A palynological assemblage from the Cambrian (Series 2, Stage 4) of Shandong Province, China, and its implications to the transition from algae to land plants. *Review of Palaeobotany and Palynology* **301**, 104645. (doi:10.1016/j.revpalbo.2022.104645)

304 40. Wellman CH, Osterloff PL, Mohiuddin U. 2003 Fragments of the earliest land plants. *Nature* **425**, 282–285. (doi:10.1038/nature01884)

305 41. Edwards D, Rogerson ECW. 1979 New records of fertile Rhyniophytina from the late Silurian of Wales. *Geological Magazine* **116**, 93–98. (doi:10.1017/s0016756800042503)

306 42. Kenrick P, Crane PR. 1997 The origin and early evolution of plants on land. *Nature* **389**, 33–39. (doi:10.1038/37918)

307 43. Silvestro D, Cascales-Miñana B, Bacon CD, Antonelli A. 2015 Revisiting the origin and diversification of vascular plants through a comprehensive Bayesian analysis of the fossil record. *New Phytologist* **207**, 425–436. (doi:10.1111/nph.13247)

308 44. Claramunt S. 2022 CladeDate : Calibration information generator for divergence time estimation. *Methods in Ecology and Evolution* **13**, 2331–2338. (doi:10.1111/2041-210x.13977)

309 45. Peters SE, Husson JM, Czaplewski J. 2018 Macrostrat: A Platform for Geological Data Integration and Deep-Time Earth Crust Research. *Geochemistry, Geophysics, Geosystems* **19**, 1393–1409. (doi:10.1029/2018gc007467)

310 46. Brown JW, Smith SA. 2017 The Past Sure is Tense: On Interpreting Phylogenetic Divergence Time Estimates. *Systematic Biology* **67**, 340–353. (doi:10.1093/sysbio/syx074)

Fig. 1.

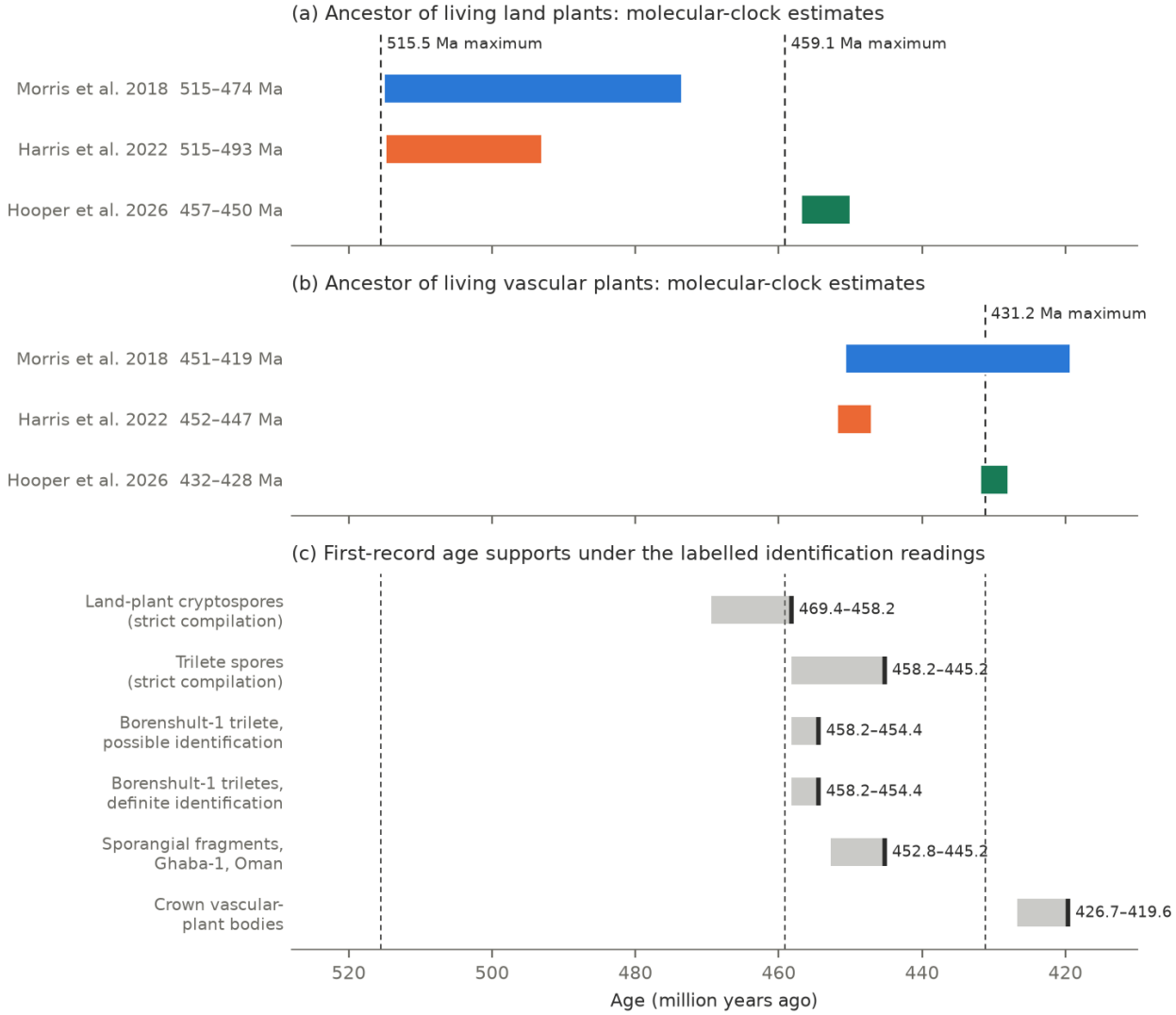


Figure 1: The dating dispute and its fossil evidence. Upper bars give dates for the ancestor of living land plants; the next group gives living vascular plants. Morris et al.'s bars envelope 95% highest posterior density (HPD) intervals across seven tree topologies; Harris et al. and Hooper et al. give their study-specific 95% HPD intervals [1–3]. Vertical lines mark the maxima of 515.5, 459.1 and 431.2 Ma. Clock estimates and maxima retain their published numerical ages. Fossil stage supports use the International Chronostratigraphic Chart v2024/12; independent radiometric ages retain their published values. Fossil bars show first-appearance supports within the declared inventories, spanning earliest possible to earliest certain occurrence; endpoints may come from different finds. Separate bars identify cryptospores and triletes under the strict compilation reading, possible and definite Borenshult triletes [21], Oman's sporangial fragments [40], and crown vascular-plant bodies [29,31]. The two Borenshult readings share this coarse age bracket. Age certainty is conditional on the labelled identification; the queried specimen remains a possible trilete.

Fig. 2.

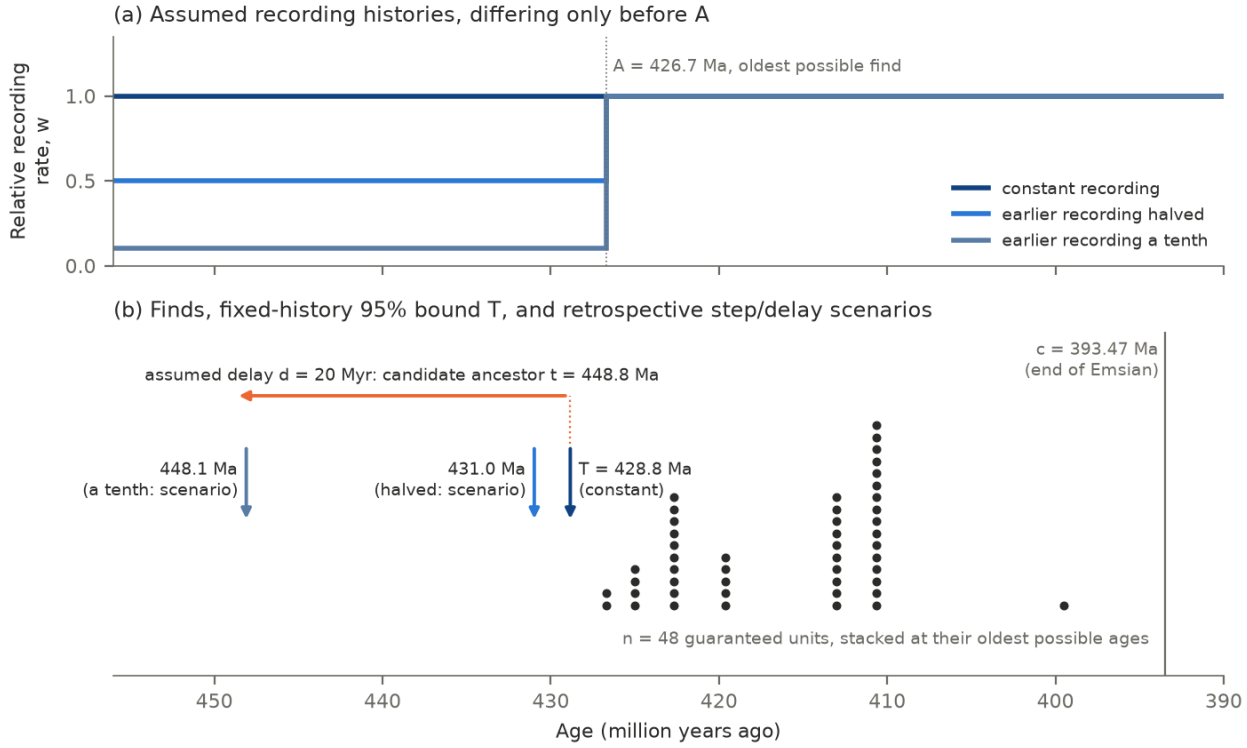


Figure 2: How the body-record calculation links fossils to ancestors, illustrated with the primary crown vascular-plant units. (*a*) Later recording is constant; earlier recording is unchanged, halved or reduced to a tenth. (*b*) Dots show guaranteed units at the older ends of their recorded age ranges, stacked for visibility. For a specified fixed recording history, the oldest find A and count n give a one-sided 95% older bound T on when the recognisable record began. For range-dated finds, A is the greatest possible oldest age and n the minimum guaranteed count. In the retrospective step scenarios, lower earlier recording moves the compatibility endpoint older; an assumed delay d before recognisable bodies connects record start to candidate ancestor age t. The selected step's curves describe retrospective compatibility: the endpoints are 431.0 Ma when earlier recording is halved and 448.1 Ma at a tenth, compared with the constant-history bound 428.8 Ma. The arrow adds an assumed 20 Myr delay to that constant-history endpoint, giving 448.8 Ma. For the chosen units and cutoff, sparser earlier recording or an assumed diagnostic delay can accommodate ancestors older than the fixed-history bound.

Fig. 3.

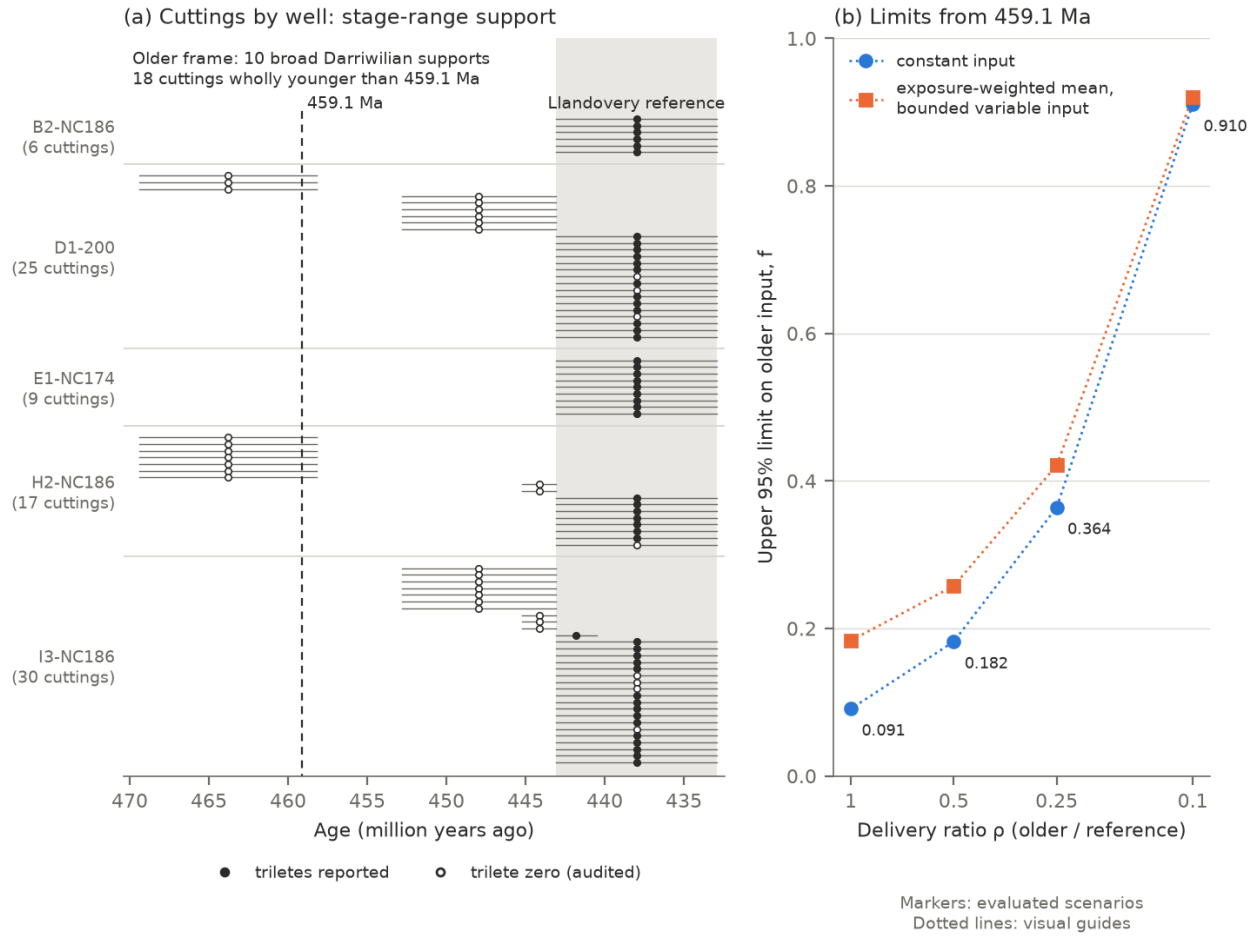


Figure 3: Spore records from the Murzuq Basin, Libya [26]. (*a*) Age ranges of chart-coded cuttings (drilling chips) from five wells. Filled symbols denote triletes; open symbols denote audited trilete-zero reports, from explicit statements or complete reporting charts. Symbols are centred on their age-support bars for display; the primary model integrates uniform ages over each support. Horizontal separators identify wells, with cutting counts beside their names. All 28 older cuttings are coded zero; 51 of 59 Llandovery (earliest Silurian) reference cuttings are positive. Of the older frame, 18 cuttings are wholly younger than 459.1 Ma; only 10 broadly dated Darriwilian cuttings in two wells have support extending older. Shading marks the reference interval; the dashed line marks 459.1 Ma. (*b*) Upper 95% compatibility limits on older input divided by reference input, f, for constant histories and the exposure-weighted mean of bounded variable histories beginning at 459.1 Ma. The older/reference delivery ratio ρ includes transport, preservation and preparation; $\rho = 1$ means equal delivery. Markers identify the four evaluated delivery scenarios; dotted connecting lines are visual guides. At a marked ratio, input below its limit remains compatible under the stated history class and primary shared-well model. Poorer delivery permits input near the reference level.

Fig. 4.

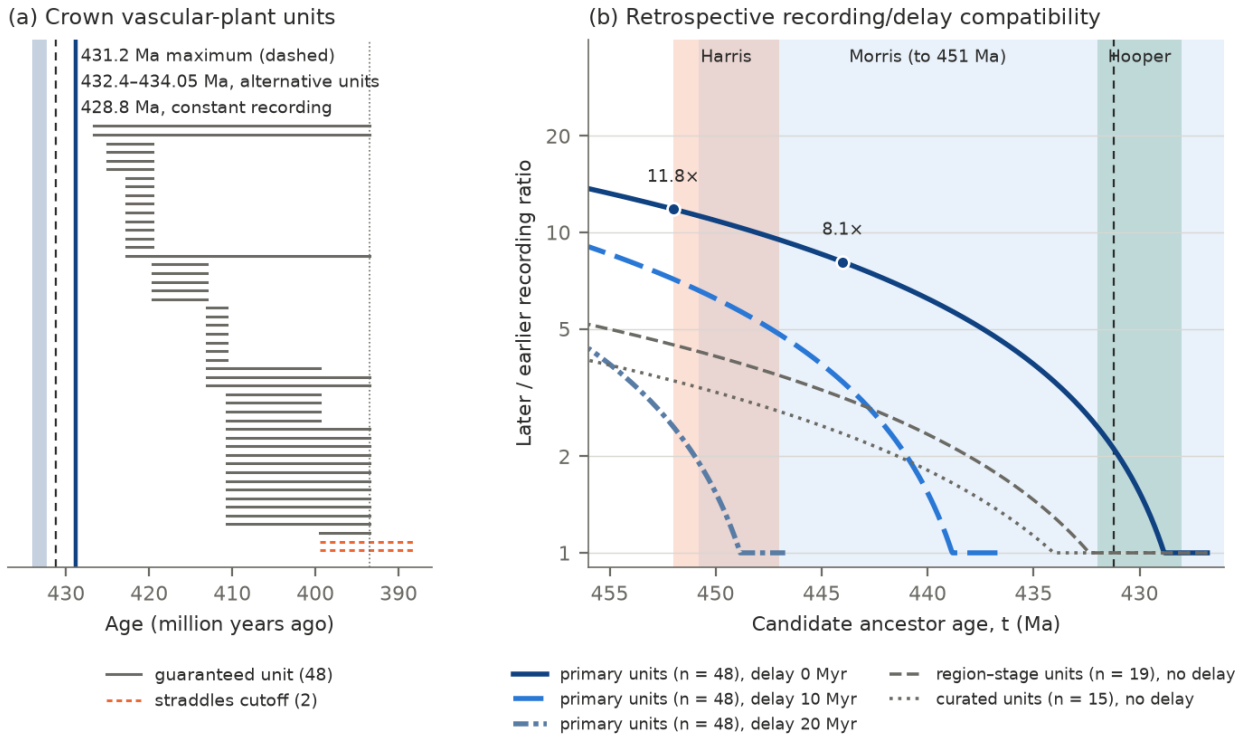


Figure 4: Recording scenarios compatible with older vascular-plant ancestors. (*a*) The 50 possible locality–age units, ordered by oldest possible age; 48 are guaranteed within the analysis window, and the two cutoff-straddling units are marked separately. The one-sided 95% older bound under constant recording is 428.8 Ma; region–stage and curated alternatives give 432.4–434.05 Ma. (*b*) Later-to-earlier recording ratios on retrospective compatibility contours. Later recording is constant, the step is at the oldest possible age $A = 426.7$ Ma, and the younger cutoff is $c = 393.47$ Ma. The contour sets the probability of an oldest recorded age at most A to 0.05, conditional on count. Primary curves use $n = 48$ and assumed diagnostic delays of zero, ten and twenty Myr; zero-delay comparison curves use region–stage ($n = 19$) and curated ($n = 15$) units. Shading identifies the published clock summaries; Morris's younger interval continues outside this panel (figure 1). Undefined combinations with $t - d \leq A$ are omitted; ratios requiring no reduction are drawn at one. These data-selected contours describe compatibility scenarios. Older ancestors fit if early bodies were recorded several times less often, or recognisable bodies appeared later.